

Mark Scheme (Results)

Summer 2007

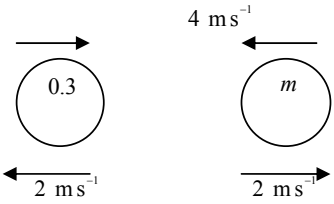
GCE

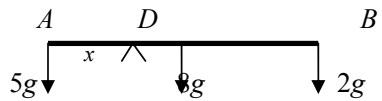
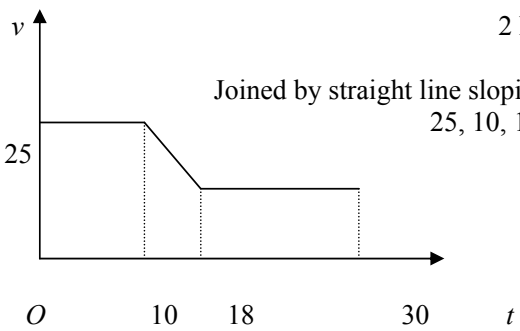
GCE Mathematics

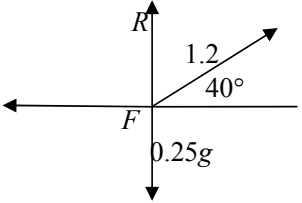
Mechanics M1 6677



June 2007
6677 Mechanics M1
Mark Scheme

Question Number	Scheme	Marks
1.	(a) $\rightarrow T \sin 20^\circ = 12$ $T \approx 35.1$ (N) awrt 35 12 T 20° (b) $\uparrow W = T \cos 20^\circ$ ≈ 33.0 (N) awrt 33	M1 A1 A1 (3) M1 A1 DM1 A1 (4) [7]
2.	 (a) $A: I = 0.3(8 + 2)$ $= 3 \text{ (Ns)}$ (b) LM $0.3 \times 8 - 4m = 0.3 \times (-2) + 2m$ $m = 0.5$ Alternative to (b) B: $m(4 + 2) = 3$ $m = 0.5$ The two parts of this question may be done in either order.	M1 A1 A1 (3) M1 A1 DM1 A1 (4) [7] M1 A1 DM1 A1 (4)

Question Number	Scheme	Marks
3.	(a) $M(C) \ 8g \times (0.9 - 0.75) = mg(1.5 - 0.9)$ Solving to $m = 2$ * (b)  $M(D) \quad 5g \times x = 8g \times (0.75 - x) + 2g(1.5 - x)$ Solving to $x = 0.6$ ($AD = 0.6$ m)	M1 A1 DM1 A1 (4) cso M1 A2(1, 0) DM1 A1 (5) [9]
4.	(a)  2 horizontal lines Joined by straight line sloping down 25, 10, 18, 30 oe (b) $25 \times 10 + \frac{1}{2}(25 + V) \times 8 + 12 \times V = 526$ Solving to $V = 11$ (c) $"v = u + at" \Rightarrow 11 = 25 - 8a$ $a = 1.75 \text{ (ms}^{-2}\text{)}$	B1 B1 B1 (3) M1 A1 A1 DM1 A1 (5) ft their V M1 A1 ft A1 (3) [11]

Question Number	Scheme	Marks
5.	(a)  $\uparrow \quad \pm R + 1.2 \sin 40^\circ = 0.25g$ Solving to $R = 1.7 \text{ (N)}$ (b) $\rightarrow \quad F = 1.2 \cos 40^\circ \text{ } (\approx 0.919)$ Use of $F = \mu R$ $1.2 \cos 40^\circ = \mu R$ $\mu \approx 0.55$	M1 A1 DM1 A1 (4) M1 A1 B1 DM1 A1ft A1 cao (6) [10]

Question Number	Scheme	Marks
6.	(a) $s = ut + \frac{1}{2}at^2 \Rightarrow 3.15 = \frac{1}{2}a \times \frac{9}{4}$ $a = 2.8 \text{ (ms}^{-2}\text{)} *$	M1 A1 A1 (3) cso
	(b) N2L for P : $0.5g - T = 0.5 \times 2.8$ $T = 3.5 \text{ (N)}$	M1 A1 A1 (3)
	(c) N2L for Q : $T - mg = 2.8m$ $m = \frac{3.5}{12.6} = \frac{5}{18} *$	M1 A1 DM1 A1 (4) cso
	(d) The acceleration of P is equal to the acceleration of Q .	B1 (1)
	(e) $v = u + at \Rightarrow v = 2.8 \times 1.5$ (or $v^2 = u^2 + 2as \Rightarrow v^2 = 2 \times 2.8 \times 3.15$) $(v^2 = 17.64, v = 4.2)$	M1 A1
	$v = u + at \Rightarrow 4.2 = -4.2 + 9.8t$ $t = \frac{6}{9.8}, 0.86, 0.857 \text{ (s)}$	DM1 A1 DM1 A1 (6) [17]

Question Number	Scheme	Marks
7.	(a) $\mathbf{v} = \frac{8\mathbf{i} + 11\mathbf{j} - (3\mathbf{i} - 4\mathbf{j})}{2.5}$ or any equivalent $\mathbf{v} = 2\mathbf{i} + 6\mathbf{j}$	M1 A1 A1 (3)
	(b) $\mathbf{b} = 3\mathbf{i} - 4\mathbf{j} + \mathbf{v}t$ ft their $\mathbf{v}$ $= 3\mathbf{i} - 4\mathbf{j} + (2\mathbf{i} + 6\mathbf{j})t$	M1 A1 ft A1cao (3)
	(c) $\mathbf{i}$ component: $-9 + 6t = 3 + 2t$ $t = 3$	M1 M1 A1
	$\mathbf{j}$ component: $20 + 3\lambda = -4 + 18$ $\lambda = -2$	M1 A1 (5)
	(d) $v_B = \sqrt{2^2 + 6^2}$ or $v_C = \sqrt{6^2 + (-2)^2}$ Both correct The speeds of B and C are the same cso	M1 A1 A1 (3) [14]